ANSWER KEY & MARKING SCHEME SET A

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** mixed without chemical combination	Components keep their properties
2	**(d)** sea water	Salt dissolved evenly
3	**(a)** copper and tin	Example of alloy
4	**(d)** the same type of particles	All particles behave identically
5	**(a)** Water	Hydrogen and oxygen chemically combined
6	**(d)** 1 : 1	Common salt
7	**(b)** hydrogen sulfide gas	A gas with a smell
8	**(c)** mixtures of minerals	Seen under magnifier
9	**(b)** magnesium oxide	A compound
10	**(d)** Fruit juice	Contains many substances

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Two or more substances mixed without chemical combination (1). Each component keeps its own properties (1).	1+1
12	Uniform: sea water, air (1). Non-uniform: muddy water, sand and iron filings (1).	1+1
13	Solid mixtures of metals with other metals or non-metals (1). Brass and bronze (1).	1+1
14	Substance with only one type of particles (1). Elements and compounds (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Components are not chemically bound (1½). Magnet separates iron from sulfur (1½).	1½+1½
16	Elements: oxygen, gold (1). Compounds: carbon dioxide, water (1). Mixtures: sea water, air (1).	1+1+1
17	Mix iron filings and sulfur (1). Heat strongly (1). Black mass iron sulfide forms and is not attracted by magnet (1).	1+1+1
	OR: Iron is magnetic, sulfur yellow (1). Iron sulfide black and not magnetic (1). It reacts with acid to give hydrogen sulfide (1).	1+1+1
18	Passing electric current splits water (1). Hydrogen and oxygen are collected (1). Shows water has two elements chemically combined (1).	1+1+1
	OR: Solid–solid: baking powder (1). Solid–liquid: sugar solution (1). Gas–gas: air (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition (1). Uniform with example (1). Non-uniform with example (1). Alloys defined (1). Three alloy examples (1).	1+1+1+1+1
	OR: Composition (1). Properties (1). Separation (1). Formation (1). Example (1).	1+1+1+1+1
20	Pure substance defined (1). Elements with examples (1). Compounds with examples (1). Chart (1). Why air is not pure (1).	1+1+1+1+1
	OR: Mixing, magnet test (1). Heating (1). Black mass formed (1). Not magnetic (1). Iron + sulfur → iron sulfide (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** components	Each retains properties
2	**(c)** muddy water	Particles settle and can be seen
3	**(b)** two or more metals	Properties different from the metals
4	**(b)** elements or compounds	Two types
5	**(a)** passing electricity (electrolysis)	A chemical change
6	**(d)** magnet	Iron is attracted
7	**(b)** hydrogen	Gas A in the figure
8	**(b)** minerals that are pure elements	Gold is an example
9	**(a)** Gold, iron, sulfur	Cannot be broken down
10	**(a)** Components keep their own properties	Contrast with compounds

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Simplest substance that cannot be broken down (1). Oxygen and gold (1).	1+1
12	Substance of two or more elements combined chemically in a fixed ratio (1). Water or sodium chloride (1).	1+1
13	Hydrogen and oxygen are combined chemically in fixed ratio (1). Properties differ from its elements (1).	1+1
14	Bring a magnet near it (1). Iron sticks, sulfur remains (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It is an alloy of iron (1). It is stronger and more durable (1). It does not rust easily (1).	1+1+1
16	Brass: copper and zinc (1). Bronze: copper and tin (1). Used for utensils, statues or bells (1).	1+1+1
17	Made of sodium and chlorine in 1 : 1 ratio (1). Evaporation is a physical process (1). Cannot split it into elements (1).	1+1+1
	OR: Element has one kind of atom; compound has two or more (1½). Element cannot be split chemically; compound can (1½).	1½+1½
18	Iron reacts with oxygen and moisture (1). New substance iron oxide forms (1). It has fixed composition (1).	1+1+1
	OR: No (1). An element has one kind of matter (1). A compound has two or more elements combined (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Passing current through water (1). Hydrogen and oxygen collected (1). Cannot be split further (1). They are elements (1). Water is a compound (1).	1+1+1+1+1
	OR: Mineral defined (1). Fixed composition (1). Rocks are mixtures of minerals (1). Native minerals are elements (1). Gold example (1).	1+1+1+1+1
20	Elements: aluminium, gold, oxygen, nitrogen, sulfur, hydrogen (2). Compounds: carbon dioxide, magnesium oxide, rust, iron sulfide, glucose, water, sodium chloride, baking soda (2). Mixtures: sand, sea water, muddy water, air, fruit juice (1).	2+2+1
	OR: Meaning (1). Examples brass, bronze, steel (1). Uses (1). Mishraloha mixtures (1). Kamsya used for health (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.